

L: Don't Ask Why 2D

Source file name: `dont.c`, `dont.cpp`, `dont.java`, or `dont.py`

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In the year 2089, the **Intergalactic Center for Protein Crafting (ICPC)** stands as humanity's most prestigious molecular engineering facility, nestled in the volcanic highlands of Iceland. Its crown jewel, *SynthCore-9*, can simulate any protein folding configuration in three-dimensional space with femtosecond precision. On a quiet Thursday morning, Dr. Aldric Novak --- ICPC's lead simulation engineer and self-proclaimed "*best coder in the northern hemisphere*" --- deployed what he described in the commit message as "*a small, totally harmless maintenance update*" to *SynthCore-9*. Within minutes, the simulator locked itself into **2D Planar Mode**: a forgotten legacy setting that forces every protein chain to fold like a path traced on an infinite grid, never crossing itself, never lifting off the plane. Before he could file an incident report, Dr. Novak noticed something unsettling: the proteins synthesized in this broken mode displayed *extraordinary* stability scores --- far beyond anything *SynthCore-9* had ever achieved in 3D. He immediately fired off a message to his colleague Dr. Sofia Reyes at **POLI** (*Protein Optimization Lab Initiative*): "*I need your brain. Right now. Don't ask why everything is 2D.*"

When Dr. Reyes arrived, she found *SynthCore-9*'s output logs scattered across three monitors. She spent the first hour quietly reading, occasionally scribbling on a whiteboard. By the second coffee, she had cracked the pattern. In 2D Planar Mode, each protein is a chain of N amino acids, each classified as either **H** (hydrophobic) or **P** (polar), with exactly B amino acids labeled H. The chain is laid out on an integer grid as a *self-avoiding walk*: starting from the origin, each amino acid occupies an adjacent cell (up, down, left, or right), never revisiting a cell. The stability of the protein is measured by its **hydrophobic contacts** --- pairs of H amino acids that are not consecutive in the chain yet find themselves occupying adjacent cells in the grid. "*More contacts, more stable. Simple,*" she said. To illustrate, she sketched the smallest interesting case on the whiteboard: a chain of $N = 4$ amino acids with a budget of $B = 2$ hydrophobic amino acids.

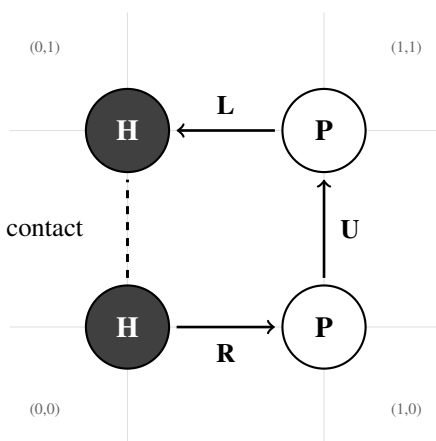


Figure 1: Sequence HPPH folded with directions RUL: 1 hydrophobic contact (dashed line). H nodes are drawn dark, P nodes light, so the figure remains legible in grayscale.

The sequence HPPH folded as RUL places the two H amino acids at opposite ends of the walk --- yet adjacent in the grid. Despite being separated by two P amino acids in the chain, they form one hydrophobic contact, the maximum achievable for these parameters. "*Someone has to find the optimal sequence and fold for every pending order,*" she said, already reaching for the keyboard.

Describing a protein. A protein is described by two strings:

- An **amino acid sequence**: a string of length N over the alphabet $\{H, P\}$ (hydrophobic / polar) with exactly B occurrences of H. The i -th character is the class of the i -th amino acid in the chain.
- A **folding**: a string of $N - 1$ *directions* that lays the chain on the grid. The first amino acid starts at the origin, and each subsequent amino acid is placed one cell away from the previous one, in the given direction. The four directions and their effect on the current cell (x, y) are:

$$R : (x+1, y) \quad L : (x-1, y) \quad U : (x, y+1) \quad D : (x, y-1)$$

The walk must be **self-avoiding**: no two amino acids may share a cell. Two amino acids are *adjacent in the grid* if their cells differ by exactly one step (one of the four directions above). A **hydrophobic contact** is a pair of H amino acids that are adjacent in the grid but *not* consecutive in the chain.

Dr. Novak's supervisor lands in Reykjavik in less than three hours. The pending synthesis queue holds several urgent orders, each arriving as a pair of integers N and B . For each order, determine the **maximum number of hydrophobic contacts** achievable, and provide an amino acid sequence and a folding that produce it --- all on a single line. The queue ends when $N = 0$ and $B = 0$. Dr. Reyes is waiting.

Input

The input consists of multiple synthesis orders. Each order is given on a single line containing two integers N and B , where

$$1 \leq N \leq 15, \quad 0 \leq B \leq N.$$

There are at most $T \leq 60$ orders before the terminating line. The input terminates when a line containing $0 \ 0$ is encountered.

The input must be read from standard input.

Output

For each order, print a single line with three space-separated tokens:

1. the maximum number of hydrophobic contacts;
2. an amino acid sequence of length N (a string of H/P with exactly B occurrences of H);
3. a folding: a string of $N - 1$ directions from $\{R, L, U, D\}$ (as defined above) that lays the sequence on the grid as a self-avoiding walk achieving that number of contacts.

If $N = 1$, print only the contact count and the sequence (there is no folding). Many optimal solutions may exist; **any** sequence and folding that achieves the maximum number of contacts is accepted.

The output must be written to standard output.

Sample Input	Sample Output
4 2	1 HPPH RUL
6 4	2 HHPPHH RRULL
3 2	0 HHP RR
1 1	0 H
0 0	